

MACMILLAN PASS, YT, Canada

WMO: 719900

Lat: 63.244N

Lon: 130.037W

Elev: 1379

StdP: 85.82

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-36.4	-34.0	-40.5	0.1	-36.4	-38.1	0.1	-34.0	11.5	-3.4	9.7	-7.1	2.2	30	0.854

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	20.7	11.2	18.5	10.2	16.6	9.5	12.0	18.9	11.0	16.9	10.2	15.3	3.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.2	8.6	12.3	8.4	8.1	11.8	7.6	7.6	11.3	37.8	18.9	35.4	16.9	33.3	15.5	15.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.4	6.4	-38.7	23.3	2.7	2.2	-40.6	24.9	-42.2	26.2	-43.7	27.4	-45.7	29.0		
(5)			-38.7	13.1	2.8	0.8	-40.7	13.7	-42.4	14.2	-43.9	14.7	-46.0	15.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.3	-17.2	-17.6	-16.5	-7.8	1.7	8.9	10.6	8.4	2.5	-5.8	-14.7	-16.8
	DBStd	12.37	8.98	7.98	8.17	5.75	4.81	3.07	2.52	3.34	3.53	5.12	7.35	7.17
	HDD10.0	5667	843	774	821	534	258	56	22	73	226	490	741	829
	HDD18.3	8618	1102	1007	1079	784	514	283	239	308	476	748	991	1088
	CDD10.0	89	0	0	0	0	2	23	41	23	1	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	13	0	0	0	0	0	7	3	4	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	2.6	2.7	2.6	2.7	2.5	2.6	2.7	2.7	2.6	2.7	2.5	2.4	2.2
(15) Precipitation	PrecAvg	434	29	23	20	16	27	50	64	57	47	39	33	29
	PrecMax	545	52	42	49	44	46	132	152	94	78	67	86	60
	PrecMin	306	10	6	4	2	9	29	29	24	8	18	12	11
	PrecStd	52	11	9	10	9	8	19	25	20	13	10	16	11
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	-0.2	3.0	4.6	17.8	24.4	22.8	23.1	14.8	7.3	2.1	-0.2
		MCWB	0.2	-1.7	0.2	0.8	7.7	12.0	12.0	12.3	8.0	4.3	0.2	-1.4
	2%	DB	-0.8	-2.7	-1.1	2.6	13.8	20.3	20.1	20.1	11.6	3.9	-0.6	-2.4
		MCWB	-1.9	-3.6	-2.8	-0.4	6.1	10.8	10.9	11.4	6.0	1.8	-1.8	-3.4
	5%	DB	-2.9	-4.5	-3.4	1.1	10.7	17.6	18.2	17.4	9.6	2.1	-3.0	-4.4
		MCWB	-3.9	-5.3	-4.5	-1.3	4.7	9.1	10.4	10.2	5.2	0.6	-4.0	-5.3
	10%	DB	-5.0	-6.9	-5.7	-0.3	8.4	15.5	16.5	15.0	7.9	0.5	-4.6	-6.8
		MCWB	-5.8	-7.6	-6.7	-2.2	3.6	8.3	10.0	9.2	4.4	-0.8	-5.4	-7.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.6	-1.6	0.5	1.4	8.1	12.9	12.8	13.0	9.0	4.6	0.0	-1.4
		MCDB	2.2	-0.4	3.0	3.9	16.4	22.3	20.0	20.6	12.9	6.7	1.2	-0.6
	2%	WB	-1.9	-3.5	-2.4	-0.1	6.5	11.3	11.8	11.9	6.8	2.2	-2.0	-3.2
		MCDB	-0.6	-2.7	-1.0	1.8	13.2	19.0	17.9	19.1	10.6	3.4	-1.0	-2.3
	5%	WB	-3.8	-5.5	-4.3	-1.1	5.1	10.1	11.1	10.8	5.6	0.6	-4.2	-5.5
		MCDB	-3.0	-4.6	-3.4	0.7	10.3	16.3	16.4	16.4	8.8	1.8	-3.4	-4.6
	10%	WB	-5.8	-7.8	-6.5	-2.1	3.8	8.9	10.4	9.7	4.6	-0.7	-5.7	-7.7
		MCDB	-5.0	-7.2	-5.5	-0.4	8.1	14.0	15.3	14.3	7.3	0.4	-5.1	-7.2
(35) Mean Daily Temperature Range	MDBR		6.9	6.9	7.5	8.2	8.5	10.8	10.3	9.7	7.6	6.5	6.4	6.6
	5% DB	MCDBR	7.1	7.4	7.2	8.1	11.2	14.2	13.7	14.0	10.8	6.9	6.7	8.5
		MCWBR	6.3	7.0	5.9	5.8	5.8	6.6	6.3	6.7	6.3	5.1	6.0	8.2
	5% WB	MCDBR	7.3	7.4	7.0	7.1	11.0	12.8	11.6	13.1	9.5	6.5	7.0	8.6
		MCWBR	6.6	6.9	5.8	5.3	5.8	6.1	5.7	6.4	5.8	4.9	6.4	8.1
(40) Clear-Sky Solar Irradiance	taub		0.160	0.218	0.237	0.287	0.310	0.327	0.341	0.318	0.274	0.230	0.161	0.151
	taud		2.041	2.264	2.278	2.161	2.230	2.405	2.386	2.484	2.534	2.319	2.005	1.920
	Ebn at Noon		547	771	889	895	903	893	867	863	846	738	511	293
	Edh at Noon		27	54	81	115	119	103	102	85	65	50	27	14
(44) All-Sky Solar Radiation	RadAvg		0.21	0.94	2.75	4.95	5.69	5.25	4.69	3.66	2.24	0.94	0.30	0.09
	RadStd		0.02	0.12	0.22	0.26	0.35	0.29	0.29	0.34	0.26	0.13	0.04	0.01

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page